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Research Proposal

Economic Complexity and Crisis Vulnerability: The Effect of ECI on GDP Growth During the 2008–2009 Global Financial Crisis

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I. Introduction and Motivation

The 2008–2009 global financial crisis triggered one of the deepest downturn in the world economy since the Great Depression. Originating in the U.S. subprime mortgage market, the crisis quickly evolved into a worldwide contraction of credit, demand, and international trade, leading to a deep, though relatively short-lived recession (IMF, 2009). Despite the global nature of the shock, the depth of the recession and the speed of recovery varied substantially across countries.

The existing literature on the real effects of the global financial crisis has highlighted a range of factors that shaped crisis exposure and recovery: pre-crisis leverages, credit booms, fiscal space and the size of automatic stabilizers, as well as financial integration and trade openness (Obstfeld & Rogoff, 2009; Reinhart & Rogoff, 2009). These contributions, however, mostly rely on quantity-type indicators (credit-to-GDP, export-to-GDP, etc.) and say relatively little about the qualitative structure of the economy.

In parallel, a rapidly growing literature on economic complexity argues that long-run development and growth are shaped by the sophistication of a country's productive capabilities, rather than by aggregate inputs alone (Hidalgo & Hausmann, 2009; Hausmann et al. 2014). The Economic Complexity Index (ECI) of Hidalgo and Hausmann (2009) measures how diversified and how rare a country's export basket is in the space of globally traded products. Complex economies tend to export a wide range of products that few other countries can make, and this pattern is interpreted as a proxy for a deep, diversified stock of productive knowledge and capabilities. Higher ECI has been shown to predict higher future growth and more sustained structural upgrading.

However, most contributions on economic complexity focus on "normal times" and long-run growth trajectories. Much less is known about what complexity implies for a country's short-

run vulnerability to global crises. On the one hand, more complex economies may be more resilient: they have richer capability sets, more diversified export structures, better institutions and more advanced financial systems, all of which could facilitate adjustment. On the other hand, complex economies are typically highly integrated into global value chains and specialized in cyclically sensitive sectors such as machinery, transport equipment and other investment- or durable-goods industries, which suffered disproportionately during the 2008-2009 trade collapse (Bems et al., 2011; Eaton et al., 2016).

II. Research question

The central research problem of this project is thus to understand whether economic complexity mitigated or amplified the real-economy impact of the 2008–2009 crisis. Does higher pre-crisis ECI make countries more resilient or more vulnerable when a large, exogenous global shock hits? This question has not been investigated before until this day to the best of our knowledge and identifying the impact of the structure of the economies in a global crisis has a crucial significance. This impact can be observed well by focusing to the 2008-2009 crisis for multiple reasons. The crisis had endogenous reasons in the United States, however, for the world it appeared as an exogenous shock, emerging with the bankruptcy of Lehman Brothers. Since the crisis occurred simultaneously in time all around the world, this period offers an intriguing environment to investigate the effects of the economic complexity in trade on the economic growth of countries.

III. Data and variables

We construct a panel database covering 94 countries over the period 2000–2015, which is sufficient to capture both the pre-crisis trajectory and the post-crisis recovery of GDP growth. The country set includes advanced, emerging and developing economies from all major world regions.

Our main explanatory variable for the analysis, the Economic Complexity Index (ECI), is derived from international trade data and obtained from the Observatory of Economic Complexity. Its computation follows the *Method of Reflections* (Hausmann & Hidalgo, 2009),

an iterative procedure used to quantify countries' economic complexity based on the structure of their exported products.

The process begins with the construction of an M_{cp} matrix, where a value of 1 indicates that country c exports product p with a revealed comparative advantage ($RCA > 1$). From this matrix, two fundamental metrics are derived: diversity, defined as $k_{c,0} = \sum_p M_{cp}$, representing the number of products exported by a country (i.e., the sum of a country's row in the matrix); and ubiquity, defined as $k_{p,0} = \sum_c M_{cp}$, representing the number of countries capable of exporting the same product (i.e., the sum of a product's column).

These measures are subsequently iterated in a process of mutual "reflection": a country's complexity is determined by the complexity of the products it exports, while the complexity of a product depends on the complexity of the countries that export it (Hausmann & Hidalgo, 2009; Hidalgo et al., 2014). After several iterations, the system converges to a stable state, which can be expressed as an eigenvalue problem. The complexity scores of countries correspond to the second principal eigenvector of the country-product network. Finally, these values are normalized (mean = 0, standard deviation = 1), yielding the trade-based ECI (Hausmann & Hidalgo, 2009; Hidalgo et al., 2014).

Macroeconomic controls are taken primarily from the World Bank's World Development Indicators and from IMF data. Our dependent variable (Economic Growth) is the annual percentage change in the value of Gross Domestic Product, calculated in constant 2015 USD, year-on-year. GDP growth is perhaps the most widely used and one of the most extensively studied macroeconomic indicators. It is closely monitored by policymakers and analysts, and it also carries symbolic significance regarding whether an economy is capable of expanding. The control variables include GDP per capita (constant 2015 USD), trade openness (the ratio of exports and imports to GDP), government debt-to-GDP, domestic credit to the private sector (as a percentage of GDP), gross capital formation (the investment rate, as a percentage of GDP), and annual CPI inflation are included to isolate the relationship between economic complexity and economic growth from other structural and macroeconomic factors known to influence growth dynamics. GDP per capita serves as a proxy for a country's level of economic development and institutional capacity, which can shape both the scope for further growth and the nature of production structures. Trade openness captures integration into global markets, a factor closely related to both growth performance and the diversification of productive

capabilities. Government debt levels may affect growth through fiscal constraints and the space available for public investment. Similarly, domestic credit to the private sector reflects financial development and access to capital, which condition the ability of firms to innovate and expand. Gross capital formation is included as a direct measure of investment activity, a key driver of productive capacity. Finally, CPI inflation controls for macroeconomic stability, as high or volatile inflation can undermine investment, weaken institutional effectiveness, and distort relative prices. Together, these variables help ensure that the estimated effect of economic complexity on GDP growth is not confounded by broader economic conditions or structural characteristics.

All variables are aligned at annual frequency. We take logs of highly skewed level variables, such as GDP per capita and trade openness, which improves the distributional properties and the interpretation of coefficients (semi-elasticities).

For the discrete high- vs low-ECI groups, countries are ranked by their 2007 ECI and assigned to the top 30 per cent (Top30) or the bottom 30 per cent (Bottom30), while the middle 40 per cent is excluded from the baseline comparison.

IV. Methodology

The empirical strategy is built around a two-way fixed-effects (FE) panel regression, which controls for time-invariant country characteristics (e.g. geography) and common global shocks. The baseline model for real GDP growth is:

$$y_{it} = \beta^T X_{it} + \delta \times Top30_i \times Crisis_t + a_i + \lambda_t + \varepsilon_{it},$$

where y_{it} is real GDP growth in country i and year t , X_{it} is the vector of macroeconomic controls, $Top30_i$ is the high-ECI indicator based on 2007 ECI, $Crisis_t$ is the crisis dummy, a_i denotes country fixed effects, λ_t year fixed effects, and ε_{it} the error term.

The coefficient of primary interest is δ , which measures the additional change in GDP growth during the crisis for high-ECI countries relative to low-ECI countries, conditional on controls and fixed effects. A negative and statistically significant δ would imply that complex economies

suffered larger short-run growth losses during the global financial crisis than less complex economies. The key identification assumption is that, in the absence of the crisis, the difference in growth between Top30 and Bottom30 countries would have followed parallel trends conditional on controls and fixed effects.

To explore dynamics and test the plausibility of parallel trends more directly, we implement an event-study specification:

$$y_{it} = \beta^T X_{it} + \sum_{\tau} \delta_{\tau} (Crisis\{t = \tau\} \times Top30_i) + a_i + \lambda_t + \varepsilon_{it},$$

where $Crisis\{t = \tau\}$ is a dummy for year τ . We focus on 2006–2010 and treat a precrisis year as the baseline. The coefficients γ_{τ} trace the Top30–Bottom30 growth gap year by year.

Robustness checks use alternative ECI timings, such as ECI measured in 2006 or the average ECI over 2005–2007, and alternative groupings such as terciles and median splits. Alternative crisis indicators are defined for robustness: one dummy equal to 1 only in 2009, capturing the trough year of the global downturn, and another equal to 1 in 2008–2010, capturing both the buildup and the immediate recovery phase. We also perform multicollinearity diagnostics to verify that controls are not excessively collinear.

V. Expected outcomes

The project is expected to reveal a link between economic complexity and crisis vulnerability. A plausible outcome is that, conditional on macroeconomic fundamentals and fixed effects, countries with higher pre-crisis ECI experienced somewhat deeper GDP growth losses at the trough of the 2008–2009 crisis, reflecting their stronger integration into global value chains and specialization in cyclically sensitive sectors. At the same time, because complexity is associated with higher long-run growth, high-ECI economies may also show a faster recovery once global trade and financial conditions improve, so that differences between high- and low-complexity countries diminish when the crisis window includes the early recovery years.

Robustness checks with alternative crisis windows, group definitions and continuous ECI×crisis interactions are expected to clarify whether this pattern is systematic across the ECI

distribution or driven by specific regions, income groups or product structures. It is also possible that the data will show weak or heterogeneous effects, or that macro-financial indicators (such as leverage or fiscal space) absorb most of the cross-country variation in crisis performance. The empirical work is therefore explicitly designed to test competing hypotheses and to identify the conditions under which complexity is associated with greater resilience or greater vulnerability.

VI. Contribution and impact

Conceptually, the project broadens the economic complexity literature by shifting part of the focus from long-run growth and structural change to short-run crisis dynamics. It asks whether the structural features captured by ECI not only predict higher income and faster growth but also shape how economies respond to large global shocks. Methodologically, it demonstrates how network-based measures of productive structure can be combined with standard panel and difference-in-differences techniques, offering a replicable framework for future work that links granular trade or production data to aggregate macroeconomic outcomes.

On the policy side, the findings will inform debates on whether “moving up the complexity ladder” can be treated as an automatic resilience strategy or whether it entails trade-offs between higher long-run growth and greater exposure to global downturns. If complexity is found to amplify crisis losses but also speed up recovery, this would support the case for complementing industrial and innovation policies with macroprudential, fiscal and social policies tailored to highly complex, export-oriented economies. If, instead, complexity appears largely orthogonal to crisis performance once macro-financial factors are controlled for, the project will still provide useful guidance by suggesting that resilience policies should target those traditional factors more directly, while complexity policies can focus primarily on long-run development.

VII. References

Most of these papers are not cited in the text. Did you need them?

- Baldwin, R. (2009). *The Great Trade Collapse: Causes, Consequences and Prospects*. London: Centre for Economic Policy Research.
- Baldwin, R., & Evenett, S. (2009). *The collapse of global trade, murky protectionism, and the crisis: Recommendations for the G20*. London: Centre for Economic Policy Research.
- Balland, P.-A., Broekel, T., Diodato, D., Giuliani, E., Hausmann, R., & Rigby, D. (Research Policy). The new paradigm of economic complexity. 2022, Volume 51, Issue 3.
- Bems, R., Johnson, R., & Yi, K.-M. (2011). Vertical Linkages and the Collapse of Global Trade. *American Economic Review*, Vol. 101, No. 3, pp. 308-12.
- Blanchard, O. (2009). *The crisis: Basic mechanisms and appropriate policies*. International Monetary Fund: IMF Working Paper No. 09/80.
- Eaton, J., Kortum, S., Neiman, B., & Romalis, J. (2016). Trade and the Global Recession. *American Economic Review*, vol. 106, no. 11, pp. 3401-38.
- Felipe, J., Kumar, U., Abdon, A., & Bacate, M. (2012). Product complexity and economic development. *Structural Change and Economic Dynamics*, Volume 23, Issue 1, March 2012, Pages 36-68.
- Gómez-Gonzalez, J. E., Uribe, J. M., & Valencia, Ó. M. (2023). Does economic complexity reduce the probability of a fiscal crisis? *World Development*.
- Hausmann, R., Hidalgo, C., Bustos, S., Coscia, M., Simoes, A., & Yildirim, M. (2014). *The Atlas of Economic Complexity: Mapping Paths to Prosperity*. MIT Press.
- Hidalgo, C., & Hausmann, R. (2009). The building blocks of economic complexity. *Proc. Natl. Acad. Sci. U.S.A.*, 106 (26) 10570-10575.
- Hristovski, G., Gockov, G., & Stojkoski, V. (2024). Multidimensional Economic Complexity and Fiscal Crises. *Faculty of Economics, University Ss. Cyril and Methodius, Skopje, North Macedonia; 2Center for Collective Learning, ANITI, Université de Toulouse & CIAS Corvinus University of Budapest*.
- IMF. (2009). *World Economic Outlook: Crisis and Recovery*. Washington DC: International Monetary Fund.
- Lane, P., & Milesi Feretti, G. M. (2012). External Adjustment and the Global Crisis. *Journal of International Economics*, vol. 88(2), pages 252-265.
- Levchenko, A., Lewis, L., & Tesar, L. (2010). The Collapse of International Trade during the 2008-09 Crisis: In Search of the Smoking Gun. *IMF Economic Review*, vol. 58(2), pages 214-253.
- Obstfeld, M., & Rogoff, K. (2009). *Global imbalances and the financial crisis: Products of common causes (CEPR Discussion Paper No. 7606)*. California: Centre for Economic Policy Research.
- Reinhart, C., & Rogoff, K. (2009). *This Time Is Different: Eight Centuries of Financial Folly*. Princeton: Princeton University Press.
- Stojkoski, V., & Kocarev, L. (2017). The Relationship Between Growth and Economic Complexity: Evidence from Southeastern and Central Europe. *MPRA Paper 77837, University Library of Munich, Germany, revised 2017*.

- Sweet, C., & Eterovic, D. (2019). Do patent rights matter? 40 years of innovation, complexity and productivity. *World Development*, Volume 115, March 2019, Pages 78-93.
- Tacchella, A., Cristelli, M., Gabrielli, A., & Caldarelli, G. (2012). A New Metrics for Countries' Fitness and Products' Complexity. *Scientific Reports*.
- Valverde-Carbonell, J. (2025). Rethinking the literature on economic complexity indexes. *Economic Analysis and Policy*, Volume 87, September 2025, Pages 123-145.
- Wooldridge, J. (2002). *Econometric Analysis of Cross Section and Panel Data* . Cambridge: The MIT Press.
- WTO. (2009. 03. 23.). WTO sees 9% global trade decline in 2009 as recession strikes. *PRESS/554 WORLD TRADE 2008, PROSPECTS FOR 2009*.
- Zhu, S., & Li, R. (2016). Economic complexity, human capital and economic growth: empirical research based on cross-country panel data. *Applied Economics*.